香港資優教育學苑
The Hong Kong Academy for Gifted Education

IMO Training 2019 - 20 Phase III

Harder Sequences & Series

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Methods of Solving Recurrence Sequence Problems

A. Finding the Sum (求和法)

This method applies to sequences of the form: $a_{n+1} = a_n + f(n)$

(Note that if f(n) is a constant, then the sequence is an A.P.)

$a_{n+1} = a_n + f(n)$, $a_n = a_{n-1} + f(n-1)$, $a_{n-1} = a_{n-2} + f(n-2)$, …$a_2 = a_1 + f(1)$

Adding up, $a_n = a_1 + \sum_{k=1}^{n-1} f(k)$.

Example 1: Find the general term of $\{a_n\}$ defined by $a_1=1$, $a_{n+1} = a_n + 2n$ for n = 1, 2, 3, …

Solution: $a_n = a_1 + \sum_{k=1}^{n-1} 2k = 1 + [2 + 4 + .. + 2(n-1)] = 1 + (2 + 2n - 2)(n - 1)/2 = n^2 - n + 1$.

Example 2: Given that $a_1=1$ and $a_n = a_{n-1} + \frac{1}{n^2-n}$, find the general term a_n.

Solution: $a_n = a_1 + \sum_{k=2}^{n} \frac{1}{(k)(k-1)} = 1 + \sum_{k=2}^{n}\left(\frac{1}{k-1} - \frac{1}{k}\right)$

$= 1+(\frac{1}{n-1} - \frac{1}{n})+(\frac{1}{n-2} - \frac{1}{n-1})+…+(\frac{1}{2} - \frac{1}{3})+(\frac{1}{1} - \frac{1}{2})$

$= 2 - \frac{1}{n}$

Example 3: A parabola divided the plane into two regions. Two parabolas can divide the plane into at most 7 regions. What is the maximum number of regions n parabolas can divide the plane into?

Solution: Note that $a_1 = 2$, $a_{n+1} = a_n + (4n+1)$

Hence $a_n = a_1 + \sum_{k=1}^{n-1} 4k + 1 = 2 + 4(n-1)(\frac{n}{2}) + n-1 = 2n^2 - n + 1$

B. Finding the Product (求積法)

This method applies to Sequences of the form: $a_{n+1} = a_n f(n)$

(Note that if f(n) is a constant, then the sequence is a Geometric Sequence)

$a_{n+1} = a_n f(n)$, $a_n = a_{n-1} f(n-1)$, $a_{n-1} = a_{n-2} f(n-2)$, …$a_2 = a_1 f(1)$

Multiplying together, $a_n = a_1 f(1)f(2)f(3)…f(n-1)$

Example 4: Given that $a_1=1$ and $a_n = \frac{a_{n-1}}{2} + 1$, find the general term a_n.

Solution: $a_n = \frac{a_{n-1}}{2} + 1$, $(a_n - 2)= \frac{a_{n-1}-2}{2} = … = \frac{a_1-2}{2^{n-1}} = -(\frac{1}{2})^{n-1}$ => $a_n = 2 - (\frac{1}{2})^{n-1}$

C. Change of Variable (換元法)

Example 5: Solve the recursive sequence $a_1=\frac{1}{3}$ and $a_{n+1} = (\frac{4}{3})a_n - 1$

Solution: Method 1: $a_{n+1} = (\frac{4}{3})a_n - 1$ and $a_n = (\frac{4}{3})a_{n-1} - 1$

Hence $(a_{n+1} - a_n) = (\frac{4}{3})(a_n - a_{n-1})$

Let $b_n = (a_{n+1} - a_n)$, it may be seen that b_n is a geometric progression. with C.R. = $\frac{4}{3}$.

Hence $a_n = a_1 + \sum_{k=1}^{n-1} b_k = \frac{1}{3} + (-\frac{5}{9} - \frac{1}{3})[1 - (\frac{4}{3})^{n-1}]/(1 - \frac{4}{3})]$

$= 3 - 8(\frac{4^{n-1}}{3^n})$

Method 2: Try, $(a_{n+1} - r) = (\frac{4}{3})(a_n - r)$, we have $r - \frac{4r}{3} = -1$, $r = 3$

So $(a_{n+1} - 3) = (\frac{4}{3})(a_n - 3)$

$a_n = 3 + (\frac{1}{3} - 3)(\frac{4}{3})^{n-1} = 3 - 8(\frac{4^{n-1}}{3^n})$

D. Solving Recurrence Sequences of the form $a_{n+1} = p\,a_n + q$, where $p \neq 1$.

Method 1: From $a_{n+1} = p\,a_n + q$, $a_n = p\,a_{n-1} + q$

We have $a_{n+1} - a_n = p(a_n - a_{n-1})$

Let $b_n = a_{n+1} - a_n$, then $b_n = p\,b_{n-1}$, which is a G.P.

Hence $a_n = a_1 + \sum_{k=1}^{n-1} b_k = a_1 + \frac{b_1(1-p^{n-1})}{1-p} = a_1 + \frac{(a_2-a_1)(1-p^{n-1})}{1-p}$

Method 2: From $a_{n+1} = p\,a_n + q$, transform to $(a_{n+1} - r) = p(a_n - r)$, we have $r = \frac{q}{1-p}$

So, $(a_{n+1} - \frac{q}{1-p}) = k(a_n - \frac{q}{1-p}))$

Continuing the process, $(a_{n+1} - \frac{q}{1-p}) = (a_1 - \frac{q}{1-p})p^n$

Hence $a_n = \frac{q}{1-p} + (a_1 - \frac{q}{1-p})p^{n-1}$

Example 6: Given that $a_1=2$ and $na_{n+1} = (n+1)a_n + 2$, find the general term a_n.

Solution: Divide both sides by $n(n+1)$, and let $b_n = \frac{a_n}{2}$, we have

$b_{n+1} = b_n + \frac{2}{n^2+n}$ and $b_1 = 2$

Hence $b_n = b_1 + 2\sum_{k=1}^{n-1}\frac{1}{k(k+1)} = 2 + 2\sum_{k=1}^{n-1}\left(\frac{1}{k} - \frac{1}{k+1}\right)$

$= 4 - \frac{2}{n}$,

and $a_n = 4n - 2$

Example 7: A circle is divided into $n(n\geq 2)$ sectors, and each sector is painted in either red, yellow or green, so that adjacent sectors are painted in different colours. In how many ways can the circle be printed?

Solution: Let a_n be the number of ways of painting, when there are n sectors.

Obviously $a_2 = 3(2) = 6$.

For $n>2$, consider the painting of the nth sector, if the chosen colour is the same as the first sector, then consider it as a_{n-1}. Hence $a_n + a_{n-1} = 3(2)^{n-1}$

$\frac{a_n}{2^n} - 1 = (-\frac{1}{2})(\frac{a_{n-1}}{2^{n-1}} - 1)$ => $a_n = 2^n + 2(-1)^n$.

E. Construction of a new recurrence sequence (構造新遞推序列)

Example 8: Find the general term of the sequence $\{a_n\}$ defined by $a_1 = 2k$, and $a_n = 2k - \frac{k^2}{a_{n-1}}$, where $k \neq 0$.

Solution: Obviously $a_n \neq k$, and we may rewrite as $a_n - k = \frac{k(a_{n-1}-k)}{a_{n-1}}$, or $\frac{1}{a_n-k} = \frac{1}{a_{n-1}-k} + \frac{1}{k}$.

Let $b_n = \frac{1}{a_n-k}$, then $b_n = b_{n-1} + \frac{1}{k}$

This is an arithmetic progression with first term $= \frac{1}{k}$ and common difference $= \frac{1}{k}$

Therefore $b_n = \frac{n}{k}$, and $a_n = k + \frac{1}{b_n} = k + \frac{k}{n}$

Exercise 9: The sequence $\{a_n\}$ is defined by $a_1 = 3$, and $a_{n+1} = (\frac{n+1}{n^3})a_n^3$ for $n \geq 1$.

Find the general term of $\{a_n\}$

Solution: Let $b_n = \frac{a_n}{n}$, then $b_{n+1} = b_n^3$

Finally, $a_n = n\,(3^{3^{n-1}})$ for $n \geq 1$.

Example 10: A sequence $\{a_n\}$ is defined by $a_n > 0$, $a_1 = 1$, and $a_n a_{n+1}^2 = 3^6$, find the general term of the sequence.

Solution: Taking $\log_3$, $2\log_3 a_{n+1} + \log_3 a_n = 6$

Let $b_n = \log_3 a_n$, then $2b_{n+1} + b_n = 6$, $b_1=0$,

$b_{n+1} = (-\frac{1}{2})(b_n) + 3$,

Try $(b_{n+1} - r) = (-\frac{1}{2})(b_n - r)$, we have $r + r/2 = 3$, hence $r=2$.

$b_{n+1} - 2 = (-\frac{1}{2})(b_n - 2)$.

Hence $b_n - 2 = (b_1 - 2)(-\frac{1}{2})^{n-1} = (-\frac{1}{2})^{n-2}$, $b_n = 2 + (-\frac{1}{2})^{n-2}$

Finally $a_n = 9 \cdot 3^{(-\frac{1}{2})^{n-2}}$.

Example 11: A sequence $\{a_n\}$ is defined by $a_1 = 0$, $a_2 = 1$, $a_3 = 2$, and $a_n = (n-1)(a_{n-1} + a_{n-2})$ for $n \geq 4$. Find the general term of the sequence.

Solution: Since for $n \geq 4$, $a_n = (n-1)(a_{n-1} + a_{n-2})$

Divide by n! to get $\frac{a_n}{n!} = (1-\frac{1}{n})(\frac{a_{n-1}}{(n-1)!}) + (\frac{1}{n})(\frac{a_{n-2}}{(n-2)!})$

Let $b_n = \frac{a_n}{n!}$, then $b_n - b_{n-1} = (-1)(\frac{b_{n-1}-b_{n-2}}{n})$, for $n \geq 4$

Hence $b_n - b_{n-1} = \frac{(-1)^{n-2}(b_2-b_1)}{n(n-1)(n-2)\ldots(4)(3)} = \frac{(-1)^n}{n!}$

and $b_{n-1} - b_{n-2} = = \frac{(-1)^{n-1}}{(n-1)!}$!, ... etc.,

up to $b_2 - b_1 = \frac{(-1)^2}{2!}$, and $b_1=0$.

Finally, $b_n = \sum_{m=2}^{n} \frac{(-1)^m}{m!}$, and $a_n = n! \sum_{m=2}^{n} \frac{(-1)^m}{m!}$

F. Roots of the Characteristic Equation (特徵根方法)

Example 12: Given $a_{n+1} = 2a_n + 3a_{n-1} - 4$, and $a_1 = 1$, $a_2 = 2$, find a_n.

Solution: The characteristic equation is $x^2 - 2x - 3 = 0$,Solving gives $x_1 = 3$ and $x_2 = -1$

Try $a_n = k_1x_1^n + k_2x_2^n + k_3$

$(k_1x_1^{n+1} + k_2x_2^{n+1} + k_3) = 2(k_1x_1^n + k_2x_2^n + k_3) + 3(k_1x_1^{n-1} + k_2x_2^{n-1} + k_3) - 4$

because x_1 and x_2 are the roots of the characteristic equation, therefore

$k_3 = 2(k_3) + 3(k_3) - 4$, giving $k_3 = 1$

Next, we try to solve for k_1 and k_2, we have $1 = 3k_1 - k_2 + 1$, and $2 = 9k_1 + k_2 + 1$

Solving gives $k_1 = \frac{1}{12}$, $k_2 = \frac{1}{4}$,

hence $a_n = (\frac{1}{12})3^n + (\frac{1}{4})(-1)^n + 1$.

Example 13: Consider the sequence characterized by $a_{n+1} = 3a_n - 2a_{n-1}$ and $a_1 = 3$, $a_2 = 4$, find a_n.

Solution: $a_{n+1} = 2a_n - a_{n-1}$, $(a_{n+1} - a_n) = 2(a_n - a_{n-1}) = \ldots = 2^{n-2}(a_2 - a_{n-1})$

Therefore $a_n - a_1 = 1 + 2 + 4 + \ldots, +2^{n-2} = 2^{n-1} - 1$

G. Making Use of Mathematical Induction (數學歸納法)

Example 14: Given that a_n is the number of possible integers N whose digits can only be 1, 3 or 4, and they add up to n. For example $a_1 = 1$, $a_2 = 1$, $a_3 = 2$, $a_4 = 4$ (here the four numbers are 1111. 13. 31 and 4). Prove that a_{2k} is a perfect square.

Proof: Consider $a_n = x_1 + x_2 + x_3 + \ldots + x_n$.

For n>4, note that x_1 can be 1, 3 or 4, hence $a_n = a_{n-1} + a_{n-3} + a_{n-4}$.

The characteristic equation is $x^4 - x^3 - x - 1 = 0$

Solving gives the roots $\pm i$, $(1 \pm \sqrt{5})/2$

Hence $a_n = k_1 i^n + k_2(-i)^n + k_3(\frac{1+\sqrt{5}}{2})^n + k_4(\frac{1-\sqrt{5}}{2}))^n$

Using $a_1 = 1$, $a_2 = 1$, $a_3 = 2$, $a_4 = 4$, we get $k_1 = \frac{2-i}{10}$, $k_2 = \frac{2+i}{10}$, $k_3 = \frac{1}{5}(\frac{1+\sqrt{5}}{2})^2$, $k_4 = \frac{1}{5}(\frac{1-+\sqrt{5}}{2})^2$.

Hence $a_n = \frac{2-i}{10} i^n + \frac{2+i}{10}(-i)^n + \frac{1}{5}(\frac{1+\sqrt{5}}{2})^{n+2} + \frac{1}{5}(\frac{1-\sqrt{5}}{2})^{n+2}$.

From above, $a_{2k} = \frac{1}{5}[(\frac{1+\sqrt{5}}{2})^{n+1} - \frac{1}{5}(\frac{1-\sqrt{5}}{2})^{n+1}]^2$, and

$a_{2k+1} = \frac{1}{5}[(-1)n + (\frac{1+\sqrt{5}}{2})^{2n+3} - \frac{1}{5}(\frac{1-\sqrt{5}}{2})^{2n+3}]$

Looking at the figures: $a_1 = 1$, $a_2 = 1$, $a_3 = 2$, $a_4 = 4$, a5 = 6, a6 = 9, a7 = 15, a8 = 25, ………

we guess that $a_{2n+2} = (a_{n+1} + a_{n-1})^2$ and $a_{2n+3} = (a_{n+1} + a_{n-1})(a_{n+2} + a_n)$

They can be proved using Mathematical Induction.

Assuming that the proposition is tru for n = k (k>4), then

$$
\begin{aligned}
a_{2k+2} &= a_{2k+1} + a_{2k-1} + a_{2k-2} \\
&= (a_{k+1} + a_{k-1})(a_k + a_{k-2}) + (a_k + a_{k-2})(a_{k-1} + a_{k-3}) + (a_{k-1} + a_{k-2})^2 \\
&= (a_{k+1} + a_{k-1})(a_k + a_{k-2}) + (a_{k-1} + a_{k-3})(a_k + a_{k-1} + a_{k-2} + a_{k-3}) \\
&= (a_{k+1} + a_{k-1})(a_k + a_{k-2} + a_{k-1} + a_{k-3}) \\
&= (a_{k+1} + a_{k-1})^2
\end{aligned}
$$

$$
\begin{aligned}
a_{2k+32} &= a_{2k+21} + a_{2k} + a_{2k-1} \\
&= (a_{k+1} + a_{k-1})^2 + (a_k + a_{k-2})^2 + (a_k + a_{k-2})(a_{k-1} + a_{k-3}) \\
&= (a_{k+1} + a_{k-1})^2 + (a_k + a_{k-2})(a_k + a_{k-2} + a_{k-1} + a_{k-3}) \\
&= (a_{k+1} + a_{k-1})^2 + (a_k + a_{k-2})(a_{k-1} + a_{k-1}) \\
&= (a_{k+1} + a_{k-1})(a_{k+2} + a_k)
\end{aligned}
$$

Using the Principle of Mathematical Induction, the proposition is true for all positive integers.

Example 15: Define $a_1 = 1$ and $a_2 = 7$ and $a_{n+2} = (a_{n+1}^2 - 1)/a_n$ for positive integer n. Prove that $9a_na_{n+1} + 1$ is a perfect square for every positive integer n.

Proof: Note that $a_1 = 1$ and $a_2 = 7$ and $a_{n+2} = (a_{n+1}^2 - 1)/a_n$, hence $a_3 = 48$

Since $a_{n+2} = (a_{n+1}^2 - 1)/a_n$ and hence $a_{n+1} = (a_n^2 - 1)/a_{n-1}$

Therefore $a_{n+2}\, a_n - a_{n+1}^2 = a_{n+1}\, a_{n-1} - a_n^2$ and $a_{n+2}\, a_n + a_n^2 = a_{n+1}\, a_{n-1} + a_{n+1}^2$

Hence $\frac{a_{n+3}+a_n}{a_{n+1}} = \frac{a_{n+1}+a_{n-1}}{a_n}$, and hence $= \ldots = \frac{a_3+a_1}{a_2} = 7$

Therefore $a_{n+2} = 7\, a_{n+1} - a_n$

To prove that $9a_na_{n+1} + 1$ is a perfect square, we check that $9a_1a_2 + 1 = 64 = (a_1 + a_2)^2$, and $9a_2a_3 + 1 = 3025 = (a_2 + a_3)^2$.

We guess that $9a_na_{n+1} + 1 = (a_n + a_{n+1})^2$ and this can be prove by M.I.

Since $9a_1a_2 + 1 = 64 = (a_1 + a_2)^2$, the proposition is true for n = 1.

Assume that it is true for n = k, i.e. $9a_ka_{k+1} + 1 = (a_k + a_{k+1})^2$

Then $9a_{k+1}a_{k+2} + 1 = 9a_{k+1}(7\, a_{k+1} - a_k) + 1$

$= 63\, a_{k+1}^2 - (a_k + a_{k+1})^2 + 2$ (using assumption)

$= 62\, a_{k+1}^2 - a_k^2 - 2\, a_ka_{k+1} + 2$

$= 64\, a_{k+1}^2 - a_k^2 - 2\, a_ka_{k+1} - 2(\, a_{k+1}^2 - 1)$

$= 64\, a_{k+1}^2 - a_k^2 - 2\, a_ka_{k+1} - 2(a_{k+2}\, a_k)$

$= 64\, a_{k+1}^2 - a_k^2 - 2\, a_ka_{k+1} - 2(a_k)\,(7\, a_{k+1} - a_k)$

$= 64\, a_{k+1}^2 - 16\, a_ka_{k+1} + a_k^2$

$= (8a_{k+1} - a_k)^2$

$= (a_{k+1} + a_{k+2})^2$, and the proposition is also true for n = k+1.

Using the Principle of Mathematical Induction, it is true for all positive integer n.

Example 16: (21st IMO 1979 No. 6)

A and E are opposite vertices of an octagon ABCDEFGH. A frog starts jumping at vertex A. If the frog is not at vertex E, it jumps to an adjacent vertex on either side. If the frog jumps to vertex E, it stops jumping. Let e_n be the number of different paths taking n steps to E. Prove that $e_{2n-1} = 0$ and $e_{2n} = \dfrac{x^{n-1} - y^{n-1}}{\sqrt{2}}$ for n = 1, 2, 3, …, where $x = 2 + \sqrt{2}$, $y = 2 - \sqrt{2}$.

Note that a path of n steps refer to a sequence of vertices $(P_o, P_1, P_2, ..., P_n)$ satisfying

(i) $P_o = A$, $P_n = E$

(ii) For each I, $0 \le i \le n-1$, P_i is different from E.

(iii) For each I, $0 \le i \le n-1$, P_i and P_{i+1} are adjacent vertices.

Solution:

First of all, the frog cannot reach E in an odd number of steps, hence $e_{2n-1} = 0$.

Next, by counting, it it easy to find that $e_1=e_2=e_3=e_5 = 5$, $e_4=2$, $e_6 = 8$.

Suppose the number of different paths taking n steps to A, B, C, …are respectively a_n, b_n, c_n, …

We also notice that by symmetry, $a_n = h_n$, $c_n = g_n$, $d_n = f_n$.

Further, $e_n = 2d_{n-1}$

$c_n = b_{n-1} + d_{n-1}$

$b_n = c_{n-1} + a_{n-1}$

$a_n = 2b_{n-1}$

$d_n = c_{n-1}$

Hence $e_n = 2d_{n-1} = 2c_{n-2} = 2(b_{n-3} + d_{n-3}) = 2(c_{n-4} + a_{n-4} + d_{n-3}) = 2(d_{n-3} + 2b_{n-5} + d_{n-3})$

$= 2(2\, d_{n-3} + 2(d_{n-3} - d_{n-5})) = 8\, d_{n-3} - 4\, d_{n-5} = 4\, e_{n-2} - 2\, e_{n-4}.$

We shall then use Mathematical Induction to prove that $e_{2n} = \dfrac{x^{n-1} - y^{n-1}}{\sqrt{2}}$.

First of all we see that it is true for n = 1, 2, 3,
Assume that the proposition is true for n = 1, 2, 3, …, k.

Then, $e_{2k+2} = 4e_{2k} - 2e_{2k-2} = 4(\dfrac{x^{k-1} - y^{k-1}}{\sqrt{2}}) - 2(\dfrac{x^{k-2} - y^{k-2}}{\sqrt{2}})$

$= (x+y)(\dfrac{x^{k-1} - y^{k-1}}{\sqrt{2}}) - (xy)(\dfrac{x^{k-2} - y^{k-2}}{\sqrt{2}})$

$= \dfrac{x^k - y^k}{\sqrt{2}}$

The proposition is also true for n=k+1.
By the Principal of Mathematical Induction, the proposition is true for all positive integer n.

H. Miscellaneous Problems

Example 17: The terms of a sequence $\{a_n\}$ satisfies the equation

$\sum_{k=1}^{n} {}^{n}C_k a_k = \dfrac{n}{n+1}$, for n=1, 2, …, find the general term a_n.

Solution: Try n = 1, 2, 3, 4, 5 and 6 we get $a_1 = \frac{1}{2}$, $a_2 = -\frac{1}{3}$, $a_3 = \frac{1}{4}$, $a_4 = -\frac{1}{5}$, $a_5 = \frac{1}{6}$, $a_6 = -\frac{1}{7}$,

hence we guess that $a_n = \frac{(-1)^n}{n+1}$.

We need only to prove that $\sum_{k=1}^{n} {}^{n}C_k \dfrac{(-1)^{k+1}}{k+1} = \dfrac{n}{n+1}$, for n=1, 2,…

Actually $\sum_{k=1}^{n} {}^{n}C_k \dfrac{(-1)^{k+1}}{k+1} = \sum_{k=1}^{n} {}^{n+1}C_{k+1} \dfrac{(-1)^{k+1}}{n+1} = \dfrac{1}{n+1} \sum_{k=1}^{n} {}^{n+1}C_{k+1} (-1)^{k+1}$

$= \dfrac{1}{n+1}\left[(1-1)^{n+1} - {}^{n+1}C_0 + {}^{n+1}C_1\right] = \dfrac{n}{n+1}$

Hence $a_n = \frac{(-1)^n}{n+1}$ satisfies the given condition.

Example 18: Given that for any positive integer n, $a_n > 0$ and $\sum_{i=1}^{n} a_i^3 = (\sum_{i=1}^{n} a_i)^2$, find a_n.

Solution: From $\sum_{i=1}^{n} a_i^3 = (\sum_{i=1}^{n} a_i)^2$ and $\sum_{i=1}^{n+1} a_i^3 = (\sum_{i=1}^{n+1} a_i)^2$

On subtracting, we have $a_{n+1}^3 = a_{n+1}(2\sum_{i=1}^{n} a_i + a_{n+1})$

Since $a_{n+1} > 0$, therefore $a_{n+1}^2 = 2\sum_{i=1}^{n} a_i + a_{n+1}$

Similarly $a_n^2 = 2\sum_{i=1}^{n-1} a_i + a_n$

Subtract again, $(a_{n+1} - a_n)(a_{n+1} + a_n) = a_n + a_{n+1}$
Since $a_n>0$, so $(a_{n+1} + a_n) > 0$, hence $(a_{n+1} - a_n) = 1$.
From $a_1^3 = a_1^2$, and $a_1>0$, we have $a_1=1$.
From $(a_{n+1} - a_n) = 1$, we have $a_n = a_{n-1} + 1 = a_{n-2} + 2 = … = a_1 + (n-1) = n$.
The final answer is $a_n = n$.